Experiment-10

Transformers in Vision

1. Aim

To understand the core principles of Vision Transformers (ViT) by exploring self-attention mechanisms and patch-based image representations, and to apply a pretrained or lightly fine-tuned ViT model on a CIFAR-10 subset, with detailed visualization of patch embeddings and self-attention maps for selected images.

2. Theory

Vision Transformers represent a paradigm shift in computer vision by replacing convolutional operations with attention-based learning mechanisms. Unlike Convolutional Neural Networks(CNN), which rely on local receptive fields, Vision Transformers model global relationships across the entire image using self-attention.

Stages in the VIT Training model:

- Stage1: Image patching
- Stage2 : Patch Embedding
- Stage3: Positional Encoding
- Stage4: Transformer Encoder Block
- Stage5: Classification token and MLP head

I. Image patching

Image patching refers to the process of partitioning an input image into a set of fixed-size, non-overlapping square patches, where each patch is subsequently flattened and embedded to form a token that serves as input to the transformer architecture.

Fig. 1. Is Illustration of image patching applied to an automobile image. The original input image is divided into 156 fixed-size, non-overlapping patches, demonstrating how the image is decomposed into smaller regions. Each patch represents a localized visual token that can be flattened and embedded before being processed by the transformer model. This figure serves as an example of the patch-based representation employed in Vision Transformer architectures.

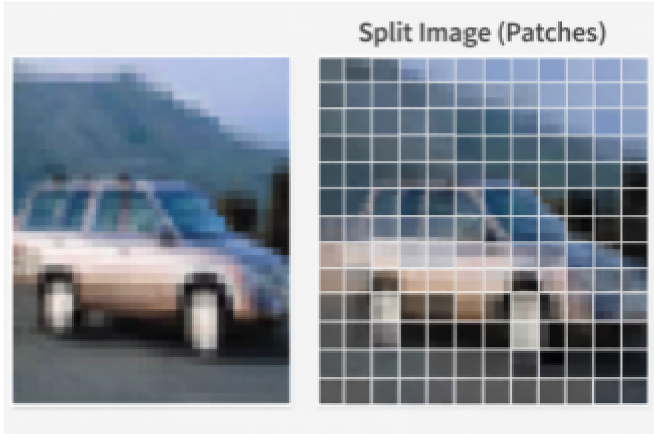


Fig 1: Image of automobile with 156 patches showing how image is converted in patches

II. Patch Embedding

In Vision Transformers, an image is first divided into fixed-size, non-overlapping patches. Each patch is flattened and linearly projected into an embedding space, forming a sequence of patch embeddings. Since transformers were originally designed for sequential data, positional embeddings are added to retain spatial information about the patches.

III. Positional Encoding in Vision Transformers

Since transformer architectures lack inherent awareness of token ordering and treat the input as an unordered set, explicit positional information must be provided. In vision-based transformers, spatial relationships are preserved by adding learnable positional embeddings to each patch token, along with the classification (CLS) token, thereby encoding the relative and absolute locations of image patches within the input sequence.

Need for Positional Encoding: Since transformer models process input tokens as an unordered set, positional encodings are incorporated to preserve spatial structure and encode patch location information within the image representation.

Common types of positional encoding used in ViT-model:

- **Learnable Positional Embeddings**: ViT uses learnable positional vectors to capture local and global spatial relationships adapting better than fixed encodings across image resolutions.

$$Z_0 = [x_{\mathrm{CLS}}\ ;\ z_1; z_2; \ldots; z_N] + E_{\mathrm{pos}}$$

- **Fixed (sinusoidal) absolute positional embeddings**
  Same idea as the original Transformer sine/cosine encoding , but applied to the patch grid positions.
  No extra learned parameters.

$$\mathrm{PE}(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{\mathrm{model}}}}}\right)$$

$$\mathrm{PE}(pos, 2i+1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{\mathrm{model}}}}}\right)$$

Where:

- `pos` = token position in the sequence (0, 1, 2, …)
- `i` = dimension index
- d_{model} = model embedding size (e.g., 512)
- Even dimensions = sine
- Odd dimensions = cosine

This makes the embedding:

- Absolute (depends on exact position)
- Deterministic (no learning required)
- Continuous & smooth.

IV. Self Attention Mechanism

The core building block of a Vision Transformer is the self-attention mechanism, which enables each patch to attend to all other patches in the image. The self-attention mechanism enables the model to capture long-range dependencies and rich contextual relationships across the input, which are often difficult to model using convolutional operations alone. Multi-head self-attention further enhances this capability by allowing the model to focus on different aspects of the image simultaneously.

Computation:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad Q = XW_Q,\ K = XW_K,\ V = XW_V$$

- Query (Q) = what this token is asking for

- Key (K) = what this token offers as information
- Value (V) = the actual content or meaning
- The attention score between tokens i and j is computed as:

$$\text{Score}(i,j) = \frac{Q_i K_j^T}{\sqrt{d_k}}$$

These scores are normalised with a softmax to produce a attention weights:

$$\alpha_{ij} = \text{softmax}_j\left(\text{Score}(i,j)\right)$$

Where: i, j = 0,1,2,3……

- QK^T computes similarity between all pairs of tokens (dot product)
- $d_k = \dfrac{D}{h}$ the dimension per attention head
- Divide by $\sqrt{d_K}$ for scaling to prevent large values causing softmax saturation
- softmax normalizes scores into probabilities for attention weights
- Multiply by V to get weighted sum of information from all tokens

V. Multi-Headed Self-Attention

Instead of a single attention operation, Vision Transformers use Multi-Head Attention so the model can capture different relationships in parallel as shown in Fig 2.
A single attention head captures only one type of relationship—perhaps syntactic, positional, or semantic. To let the model learn multiple perspectives simultaneously, the Transformer employs multi head attention (MHA).

$$\text{MultiHead}(Q,K,V) = \text{Concat}(\text{head}_1,\ldots,\text{head}_h)\,W_O$$

Residual Connections and Layer Normalization

Ensures stable training in deep networks by preserving information and normalizing activations.

- **Residual (Skip) Connections:** Residual connections bypass transformation blocks to preserve earlier layer information, preventing degradation in deep networks. They enable the model to learn incremental refinements, improving convergence and stability in deep ViTs.
- **Layer Normalization:** LayerNorm normalizes features across the input, stabilizing training and reducing internal covariate shift. Pre-LN ensures well-conditioned gradients and consistent scaling across tokens in deep Transformers.

VI. Multi-headed-attention architecture:

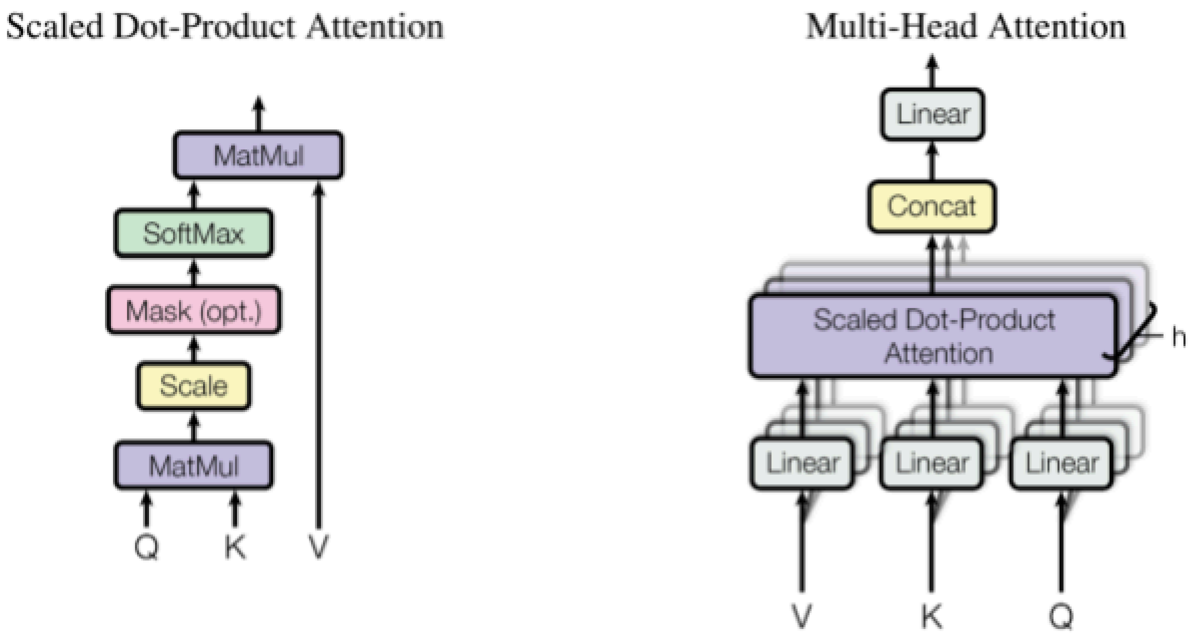


Fig 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several
attention layers running in parallel.
Source:(Vaswani *et al.*, “*Attention Is All You Need* “(2017)).

Refer to the right side of Fig .2 , where:

- The input tokens are first projected into three separate vectors—Query (Q), Key (K), and Value (V)—using learned linear transformations.
- These Q, K, and V representations are divided into *h* parallel attention heads, enabling the model to attend to information from multiple representation subspaces simultaneously.
- Within each head, scaled dot-product attention is computed as:

$$\mathrm{Attention}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where *d* denotes the dimensionality of the key vectors.

- This operation produces attention weights that quantify the relevance of each token with respect to all others, followed by a weighted aggregation of the value vectors.
- The outputs from all attention heads are concatenated and passed through a final linear projection layer to combine information from different heads.
- This mechanism enables the model to learn global contextual relationships and long-range dependencies, resulting in richer feature representations than those obtained using single-head attention or convolutional operations alone.

VII. Classification Token

A learnable classification (CLS) token is prepended to the sequence of patch embeddings to aggregate global contextual information across the input image. After propagation through the transformer encoder layers, the final CLS token representation is utilized for image-level classification. Owing to the large data requirements of Vision Transformers, pretrained models are typically adopted and subsequently fine-tuned on smaller datasets to achieve efficient learning and improved performance.

Vision Transformers typically require large datasets for effective training; therefore, pretrained models are often used and fine-tuned for smaller datasets such as CIFAR-10

VIII. Model Architecture

Encoder and Decoder Stacks:

Encoder: The encoder is composed of a stack of N = 6 as shown on the left side of Fig.3 identical layers. Each layer has two sub-layers. The first is a multi-head self-attention mechanism, and the second is a simple, position-wise fully connected feed-forward network. We employ a residual connection around each of the two sub-layers, followed by layer normalization . That is, the output of each sub-layer is LayerNorm(x + Sublayer(x)), where Sublayer(x) is the function implemented by the sub-layer itself. To facilitate these residual connections, all sub-layers in the model, as well as the embedding layers, produce outputs of dimension model = 512.

Decoder: The decoder is also composed of a stack of N = 6 as shown on the right side of Fig.3 identical layers. In addition to the two sub-layers in each encoder layer, the decoder inserts a third sub-layer, which performs multi-head attention over the output of the encoder stack. Similar to the encoder, we employ residual connections around each of the sub-layers, followed by layer normalization. We also modify the self-attention sub-layer in the decoder stack to prevent positions from attending to subsequent positions. This masking, combined with the fact that the output embeddings are offset by one position, ensures that the predictions for position i can depend only on the known outputs at positions less than i.

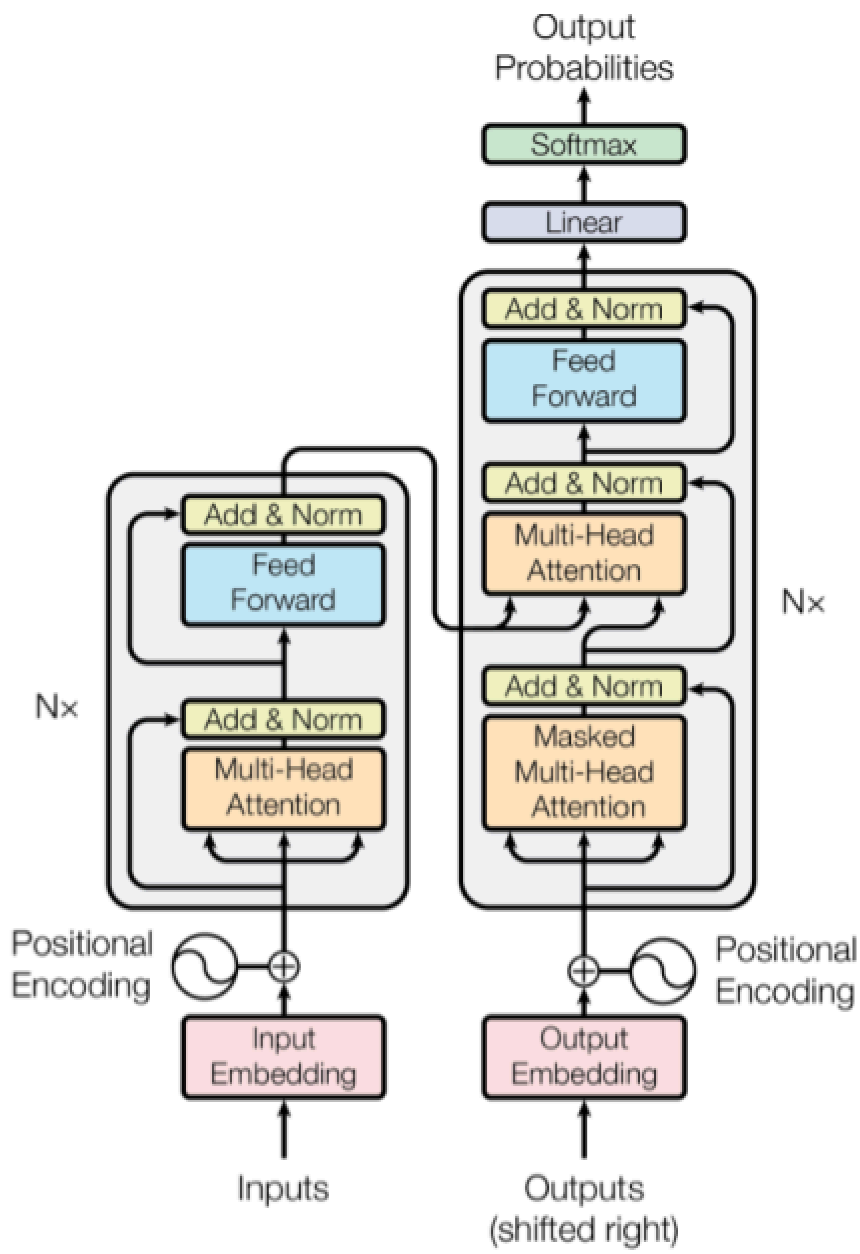


Fig 3: The Transformer - model architecture.Figure adapted from Source:(Vaswani *et al.*, "*Attention Is All You Need* ",(2017).)

IX. Transformer Encoder Block

Residual connections stabilize training, while the MLP refines learned representations. minimal content and write the equation in box

Let Z denote the input token embeddings to a transformer encoder block. The intermediate representation after multi-head self-attention and residual normalization is computed as

$$Z' = \mathrm{LayerNorm}\left(Z + \mathrm{MSA}(Z)\right)$$

Subsequently, the output of the encoder block is obtained by applying a position-wise multilayer perceptron followed by another residual connection and Layer Normalization:

$$Z_{\mathrm{out}} = \mathrm{LayerNorm}\left(Z' + \mathrm{MLP}(Z')\right)$$

Above mentioned equations together describe the standard transformer encoder structure, where residual connections preserve input information and Layer Normalization stabilizes training, while the MSA and MLP modules respectively model global dependencies and enhance feature representations.

The transformer has two parts, the decoder which is on the left side in Fig 4. and the encoder which is on the right.
Imagine we are doing machine translation for now.
The encoder takes the input data (sentence), and produces an intermediate representation of the input.
The decoder decodes this intermediate representation step by step and generates the output.

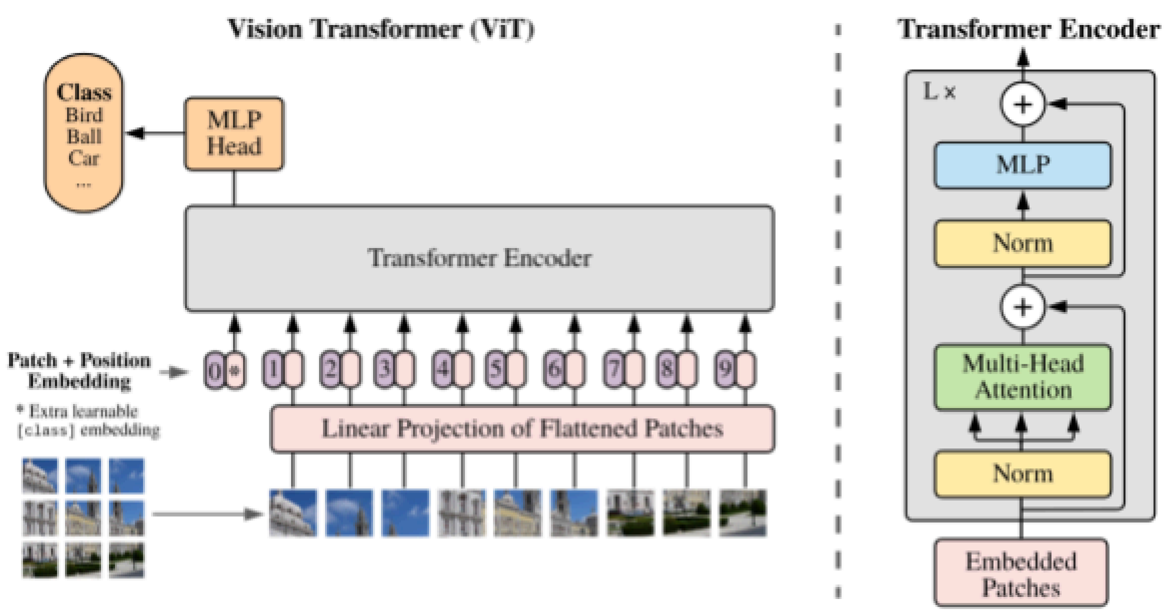


Fig 4: Model overview. We split an image into fixed-size patches, linearly embed each of them,add position embeddings, and feed the resulting sequence of vectors to a standard Transformer encoder. In order to perform classification, we use the standard approach of adding an extra learnable"classification token" to the sequence.
Source:(Vaswani et al, "An Image is worth 16x16 words"arXiv:2010.11929v2,(2017)).

Working of Encoder:

- Input (Embedded patches): Split the image into small patches and turn each patch into a vector (a token).
- Stack L times: The same encoder layer is repeated L times to improve features.

One encoder layer:

- Layer Normalization is applied before multi-head self-attention to stabilize feature distributions.
- Multi-head self-attention models global relationships among patch tokens.
- A residual connection adds the input back to preserve information and improve training stability.
- Layer Normalization followed by an MLP refines each token independently.
- A second residual connection produces enhanced token representations for downstream tasks.

X. Summary of Core Components

Component	**Purpose**	**Key Insight**
Input Embedding	Converts tokens into numerical vectors.	Enables continuous-space representation.
Positional Encoding	Adds order information to embeddings.	Introduces sequence awareness.
Self-Attention	Captures relationships between all tokens.	Enables global contextual understanding.
Multi-Head Attention	Learn multiple relation types in parallel.	Improves diversity and expressiveness.
Feed-Forward Network	Applies nonlinear transformation per token.	Enhances representation depth.
Residual + LN	Stabilizes training and gradients.	Ensures smooth optimization and deep stacking.

XI. Use Cases of Vision Transformers:

- **Access to large-scale labeled datasets and robust compute infrastructure -** vision transformers are data-hungry and require significant training time and memory, especially in their vanilla form. With enough data and compute, they are capable of outperforming CNNs in many benchmarks.

- **We need to capture long-range spatial relationships -** Unlike CNNs, which are local in their processing (remember the receptive field section), ViTs leverage self-attention to model relationships between all image patches, making them particularly useful for tasks where spatial context across the entire image matters.

- **To use pretrained models and transfer learning -** We can access pretrained ViTs (such as through Hugging Face or timm) and so fine-tuning becomes much more practical. In this case, even mid-sized datasets can give us great results without training from scratch.

Merits of Vision Transformers:

- Ability to model global context using self-attention

- Flexible architecture independent of image resolution

- Strong performance when pretrained on large datasets

- Interpretability through attention visualization

Demerits of Vision Transformers:

- High computational and memory requirements
- Large data dependency for training from scratch
- Slower convergence compared to CNNs on small datasets

3.Code and Result

1 STEP 1: Import Libraries

[1]:
```
# Core
import os
import random
from pathlib import Path
import numpy as np

# PyTorch
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, Subset

# Vision / Data
import torchvision
import torchvision.transforms as transforms
from torchvision.datasets import CIFAR10

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns

# Metrics
from sklearn.metrics import confusion_matrix, accuracy_score

# Model zoo
import timm

print("All libraries imported successfully.")
```

```
All libraries imported successfully.
```

2 STEP 2: Set Device (CPU / GPU)

[2]:
```
DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")

print(f"Using device: {DEVICE}")
if DEVICE.type == "cuda":
    print(f"GPU: {torch.cuda.get_device_name(0)}")
```

```
Using device: cuda
GPU: NVIDIA GeForce RTX 5090
```

3 STEP 3: Define Class Names

[3]:
```
CLASSES = (
    "airplane", "automobile", "bird", "cat", "deer",
    "dog", "frog", "horse", "ship", "truck"
)

NUM_CLASSES = len(CLASSES)
print(f"Number of classes: {NUM_CLASSES}")
for i, name in enumerate(CLASSES):
    print(f" {i} → {name}")
```

```
Number of classes: 10
 0 → airplane
 1 → automobile
 2 → bird
 3 → cat
 4 → deer
 5 → dog
 6 → frog
 7 → horse
 8 → ship
 9 → truck
```

4 STEP 4: Load Raw CIFAR-10 Dataset (No Transforms)

[4]:
```
DATA_ROOT = "./data"

train_dataset_raw = CIFAR10(
    root=DATA_ROOT,
    train=True,
    download=True,
    transform=None
)

test_dataset_raw = CIFAR10(
```

```
    root=DATA_ROOT,
    train=False,
    download=True,
    transform=None
)

print(f"Full training samples: {len(train_dataset_raw)}")
print(f"Full test samples: {len(test_dataset_raw)}")
```

```
Full training samples: 50000
Full test samples: 10000
```

5 STEP 5: Visualize Sample Images

[5]:
```
def show_samples(dataset, classes, n=10):
    plt.figure(figsize=(15, 4))
    for i in range(n):
        img, label = dataset[i]
        plt.subplot(1, n, i + 1)
        plt.imshow(img)
        plt.title(classes[label])
        plt.axis("off")
    plt.show()

show_samples(train_dataset_raw, CLASSES, n=10)
```

6 STEP 6: Check Class Distribution (Raw Dataset)

[6]:
```
from collections import Counter

train_labels = [label for _, label in train_dataset_raw]
label_counts = Counter(train_labels)

for cls_idx, count in label_counts.items():
    print(f"{CLASSES[cls_idx]:10s}: {count}")
```

```
frog      : 5000
truck     : 5000
deer      : 5000
```

```
automobile: 5000
bird      : 5000
horse     : 5000
ship      : 5000
cat       : 5000
dog       : 5000
airplane  : 5000
```

7 STEP 7: Define Normalization Values

[7]:
```
CIFAR_MEAN = (0.4914, 0.4822, 0.4465)
CIFAR_STD  = (0.2023, 0.1994, 0.2010)

print("Normalization values defined.")
```

```
Normalization values defined.
```

8 STEP 8: Define Strong Data Augmentation

[8]:
```
train_transform = transforms.Compose([
    transforms.RandomCrop(32, padding=4),
    transforms.RandomHorizontalFlip(),
    transforms.ToTensor(),
    transforms.Normalize(CIFAR_MEAN, CIFAR_STD),
])

test_transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize(CIFAR_MEAN, CIFAR_STD),
])
```

9 STEP 9: Visualize Augmentation Effects

[9]:
```
def show_augmented_samples(dataset, transform, classes, n=8):
    plt.figure(figsize=(15, 4))
    for i in range(n):
        img, label = dataset[i]
        img_aug = transform(img)
        img_aug = img_aug.permute(1, 2, 0)
        img_aug = img_aug * torch.tensor(CIFAR_STD) + torch.tensor(CIFAR_MEAN)
        img_aug = img_aug.clamp(0, 1)

        plt.subplot(1, n, i + 1)
        plt.imshow(img_aug)
        plt.title(classes[label])
        plt.axis("off")
```

```python
    plt.show()

show_augmented_samples(train_dataset_raw, train_transform, CLASSES)
```

10 STEP 10: Create Class-Balanced Stratified Subset + DataLoaders

Fix Random Seed (Reproducibility)

[10]:
```python
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
```

[10]:
```
<torch._C.Generator at 0x71dafc07ba50>
```

Define Subset Sizes (Per Class)

[11]:
```python
TRAIN_PER_CLASS = 2000
VAL_PER_CLASS   = 500
TEST_PER_CLASS  = 500
```

Stratified Sampling Function

[12]:
```python
def stratified_split(dataset, train_n, val_n, test_n):
    class_indices = {i: [] for i in range(NUM_CLASSES)}

    for idx, (_, label) in enumerate(dataset):
        class_indices[label].append(idx)

    train_idx, val_idx, test_idx = [], [], []

    for cls, indices in class_indices.items():
        random.shuffle(indices)
        train_idx.extend(indices[:train_n])
        val_idx.extend(indices[train_n:train_n + val_n])
        test_idx.extend(indices[train_n + val_n:train_n + val_n + test_n])

    return train_idx, val_idx, test_idx
```

Create Subset Indices

[13]:
```
train_idx, val_idx, subset_test_idx = stratified_split(
    train_dataset_raw,
    TRAIN_PER_CLASS,
    VAL_PER_CLASS,
    TEST_PER_CLASS
)

print(f"Train subset size: {len(train_idx)}")
print(f"Val subset size: {len(val_idx)}")
print(f"Subset test size: {len(subset_test_idx)}")
```

```
Train subset size: 20000
Val subset size: 5000
Subset test size: 5000
```

Create Dataset Objects with Transforms

[14]:
```
print("\n" + "="*70)
print("CREATING DATASET SPLITS")
print("="*70)

# -----------------------------
# Training Dataset
# -----------------------------
train_dataset = Subset(
    CIFAR10(root=DATA_ROOT, train=True, transform=train_transform),
    train_idx
)

print("\n[TRAIN DATASET]")
print(f"• Source           : CIFAR-10 (train split)")
print(f"• Samples          : {len(train_dataset)}")
print(f"• Augmentation     : ENABLED")
print(f"• Purpose          : Model learning")

# -----------------------------
# Validation Dataset
# -----------------------------
val_dataset = Subset(
    CIFAR10(root=DATA_ROOT, train=True, transform=test_transform),
    val_idx
)

print("\n[VALIDATION DATASET]")
print(f"• Source           : CIFAR-10 (train split)")
print(f"• Samples          : {len(val_dataset)}")
print(f"• Augmentation     : DISABLED (only normalization)")
print(f"• Purpose          : Early stopping & hyperparameter tuning")
```

```python
# ----------------------------
# Subset Test Dataset
# ----------------------------
subset_test_dataset = Subset(
    CIFAR10(root=DATA_ROOT, train=True, transform=test_transform),
    subset_test_idx
)

print("\n[SUBSET TEST DATASET]")
print(f"• Source           : CIFAR-10 (train split)")
print(f"• Samples          : {len(subset_test_dataset)}")
print(f"• Augmentation     : DISABLED (only normalization)")
print(f"• Purpose          : Fair evaluation on balanced subset")

# ----------------------------
# Full CIFAR-10 Test Dataset
# ----------------------------
full_test_dataset = CIFAR10(
    root=DATA_ROOT,
    train=False,
    transform=test_transform
)

print("\n[FULL TEST DATASET]")
print(f"• Source           : CIFAR-10 (official test split)")
print(f"• Samples          : {len(full_test_dataset)}")
print(f"• Augmentation     : DISABLED (only normalization)")
print(f"• Purpose          : Final generalization evaluation")

print("\n" + "="*70)
print("DATASET SETUP COMPLETE")
print("="*70)
```

```
======================================================================
CREATING DATASET SPLITS
======================================================================


[TRAIN DATASET]
• Source           : CIFAR-10 (train split)
• Samples          : 20000
• Augmentation     : ENABLED
• Purpose          : Model learning

[VALIDATION DATASET]
• Source           : CIFAR-10 (train split)
```

```
• Samples            : 5000
• Augmentation       : DISABLED (only normalization)
• Purpose            : Early stopping & hyperparameter tuning

[SUBSET TEST DATASET]
• Source             : CIFAR-10 (train split)
• Samples            : 5000
• Augmentation       : DISABLED (only normalization)
• Purpose            : Fair evaluation on balanced subset

[FULL TEST DATASET]
• Source             : CIFAR-10 (official test split)
• Samples            : 10000
• Augmentation       : DISABLED (only normalization)
• Purpose            : Final generalization evaluation

======================================================================
DATASET SETUP COMPLETE
======================================================================
```

11 Create DataLoaders

[15]:
```python
print("\n" + "="*70)
print("CREATING DATALOADERS")
print("="*70)

# -----------------------------
# DataLoader Configuration
# -----------------------------
BATCH_SIZE = 128
NUM_WORKERS = 4

print("\n[DATALOADER SETTINGS]")
print(f"• Batch size        : {BATCH_SIZE}")
print(f"• Num workers       : {NUM_WORKERS}")
print(f"• Pin memory        : Enabled (faster GPU transfer)")

# -----------------------------
# Training DataLoader
# -----------------------------
train_loader = DataLoader(
    train_dataset,
    batch_size=BATCH_SIZE,
    shuffle=True,
    num_workers=NUM_WORKERS,
    pin_memory=True
)
```

```python
print("\n[TRAIN DATALOADER]")
print(f"• Total samples     : {len(train_dataset)}")
print(f"• Total batches     : {len(train_loader)}")
print(f"• Shuffling         : Enabled")
print(f"• Purpose           : Model learning (each epoch sees data in new␣
↪order)")

# ------------------------------
# Validation DataLoader
# ------------------------------
val_loader = DataLoader(
    val_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=NUM_WORKERS,
    pin_memory=True
)

print("\n[VALIDATION DATALOADER]")
print(f"• Total samples     : {len(val_dataset)}")
print(f"• Total batches     : {len(val_loader)}")
print(f"• Shuffling         : Disabled")
print(f"• Purpose           : Stable validation & early stopping")

# ------------------------------
# Subset Test DataLoader
# ------------------------------
subset_test_loader = DataLoader(
    subset_test_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=NUM_WORKERS,
    pin_memory=True
)

print("\n[SUBSET TEST DATALOADER]")
print(f"• Total samples     : {len(subset_test_dataset)}")
print(f"• Total batches     : {len(subset_test_loader)}")
print(f"• Shuffling         : Disabled")
print(f"• Purpose           : Fair evaluation on balanced subset")

# ------------------------------
# Full CIFAR-10 Test DataLoader
# ------------------------------
full_test_loader = DataLoader(
    full_test_dataset,
```

```python
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=NUM_WORKERS,
    pin_memory=True
)

print("\n[FULL TEST DATALOADER]")
print(f"• Total samples     : {len(full_test_dataset)}")
print(f"• Total batches     : {len(full_test_loader)}")
print(f"• Shuffling         : Disabled")
print(f"• Purpose           : Final generalization evaluation")

print("\n" + "="*70)
print("DATALOADERS READY")
print("="*70)
```

```
======================================================================
CREATING DATALOADERS
======================================================================

[DATALOADER SETTINGS]
• Batch size         : 128
• Num workers        : 4
• Pin memory         : Enabled (faster GPU transfer)

[TRAIN DATALOADER]
• Total samples      : 20000
• Total batches      : 157
• Shuffling          : Enabled
• Purpose            : Model learning (each epoch sees data in new order)

[VALIDATION DATALOADER]
• Total samples      : 5000
• Total batches      : 40
• Shuffling          : Disabled
• Purpose            : Stable validation & early stopping

[SUBSET TEST DATALOADER]
• Total samples      : 5000
• Total batches      : 40
• Shuffling          : Disabled
• Purpose            : Fair evaluation on balanced subset

[FULL TEST DATALOADER]
• Total samples      : 10000
• Total batches      : 79
• Shuffling          : Disabled
```

• Purpose   : Final generalization evaluation

```
======================================================================
DATALOADERS READY
======================================================================
```

[16]:
```
print("\nSANITY CHECK: ONE TRAINING BATCH")

images, labels = next(iter(train_loader))
print(f"• Image batch shape : {images.shape}")
print(f"• Label batch shape : {labels.shape}")
print("Expected image shape: (batch_size, 3, 32, 32)")
print("Expected label shape: (batch_size,)")

if images.shape[1:] == (3, 32, 32):
    print("Image tensor shape is correct")
else:
    print("Image tensor shape is incorrect")
```

```
SANITY CHECK: ONE TRAINING BATCH
• Image batch shape : torch.Size([128, 3, 32, 32])
• Label batch shape : torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Expected label shape: (batch_size,)
Image tensor shape is correct
```

12 STEP 11: Build the Pretrained ViT-Small-Patch8 Model

**What this cell does (student view)

Loads a Vision Transformer pretrained on ImageNet and prepares it for CIFAR-10 classification.**

[17]:
```
MODEL_NAME = "vit_small_patch8_224"
```

[18]:
```
print("\n" + "="*80)
print("STEP 11 (FIXED): BUILDING PRETRAINED ViT-SMALL FOR CIFAR-10")
print("="*80)


model = timm.create_model(
    MODEL_NAME,
    pretrained=True,
    num_classes=NUM_CLASSES,
    img_size=32
)

model = model.to(DEVICE)
```

```
print("\n[MODEL LOADING CONFIRMATION]")
print(f"• Model name            : {MODEL_NAME}")
print("• Pretrained weights   : ImageNet")
print("• Image size override : 32×32 (CIFAR-10)")
print("• Patch size           : 8×8")
print("• Number of classes    : 10")
print(f"• Model device         : {DEVICE}")
```

```
================================================================================
STEP 11 (FIXED): BUILDING PRETRAINED ViT-SMALL FOR CIFAR-10
================================================================================


[MODEL LOADING CONFIRMATION]
• Model name            : vit_small_patch8_224
• Pretrained weights   : ImageNet
• Image size override : 32×32 (CIFAR-10)
• Patch size           : 8×8
• Number of classes    : 10
• Model device         : cuda
```

[19]:

```
print("\n" + "="*70)
print("VERIFYING MODEL INPUT COMPATIBILITY")
print("="*70)

dummy_input = torch.randn(1, 3, 32, 32).to(DEVICE)

try:
    dummy_output = model(dummy_input)
    print("  Model successfully accepted 32×32 input")
    print(f"• Output shape: {dummy_output.shape}")
    print("Expected shape: (1, 10)")
except Exception as e:
    print("  Model failed on 32×32 input")
    print(e)
```

```
======================================================================
VERIFYING MODEL INPUT COMPATIBILITY
======================================================================
  Model successfully accepted 32×32 input
• Output shape: torch.Size([1, 10])
Expected shape: (1, 10)
```

13 Verify Patch Embedding Details (VERY IMPORTANT)

[20]:
```
print("\n[PATCH EMBEDDING VERIFICATION]")

patch_size = model.patch_embed.patch_size
embed_dim = model.embed_dim

print(f"• Patch size from model      : {patch_size}")
print(f"• Embedding dimension        : {embed_dim}")

expected_patches = (32 // patch_size[0]) ** 2
print(f"• Expected patches per image : {expected_patches}")
print("• Expected tokens (+ CLS)    :", expected_patches + 1)

print("\nExplanation:")
print("• CIFAR-10 image size = 32×32")
print("• Patch size = 8×8 → (32/8)² = 16 patches")
print("• +1 CLS token → total 17 tokens")

if expected_patches == 16:
    print(" Patch-to-token mapping is correct")
else:
    print(" Patch-to-token mapping is incorrect")
```

```
[PATCH EMBEDDING VERIFICATION]
• Patch size from model      : (8, 8)
• Embedding dimension        : 384
• Expected patches per image : 16
• Expected tokens (+ CLS)    : 17

Explanation:
• CIFAR-10 image size = 32×32
• Patch size = 8×8 → (32/8)² = 16 patches
• +1 CLS token → total 17 tokens
 Patch-to-token mapping is correct
```

14 Verify CLS Token Exists

The CLS token learns a global image representation used for classification.

[21]:
```
print("\n[CLS TOKEN CHECK]")

if hasattr(model, "cls_token"):
    print(" CLS token is present in the model")
    print(f"CLS token shape: {model.cls_token.shape}")
else:
```

```
    print(" CLS token is missing (this should NOT happen)")
```

```
[CLS TOKEN CHECK]
 CLS token is present in the model
CLS token shape: torch.Size([1, 1, 384])
```

15 STEP 12: Count Model Parameters

[22]:
```
print("\n" + "="*80)
print("STEP 12: COUNTING MODEL PARAMETERS")
print("="*80)

total_params = sum(p.numel() for p in model.parameters())
trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad)

print(f"• Total parameters      : {total_params:,}")
print(f"• Trainable parameters  : {trainable_params:,}")

print("\nExpected:")
print("• ViT-Small   22 million parameters")

if total_params > 20_000_000:
    print(" Parameter count matches ViT-Small")
else:
    print("Unexpected parameter count")
```

```
================================================================================
STEP 12: COUNTING MODEL PARAMETERS
================================================================================
• Total parameters      : 21,379,210
• Trainable parameters  : 21,379,210

Expected:
• ViT-Small   22 million parameters
 Parameter count matches ViT-Small
```

16 STEP 13: Print Model Architecture

[23]:
```
print("\n" + "="*80)
print("STEP 13: MODEL ARCHITECTURE OVERVIEW")
print("="*80)

print(model)
```

```
================================================================================
STEP 13: MODEL ARCHITECTURE OVERVIEW
================================================================================
VisionTransformer(
  (patch_embed): PatchEmbed(
    (proj): Conv2d(3, 384, kernel_size=(8, 8), stride=(8, 8))
    (norm): Identity()
  )
  (pos_drop): Dropout(p=0.0, inplace=False)
  (patch_drop): Identity()
  (norm_pre): Identity()
  (blocks): Sequential(
    (0): Block(
      (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
      (attn): Attention(
        (qkv): Linear(in_features=384, out_features=1152, bias=True)
        (q_norm): Identity()
        (k_norm): Identity()
        (attn_drop): Dropout(p=0.0, inplace=False)
        (norm): Identity()
        (proj): Linear(in_features=384, out_features=384, bias=True)
        (proj_drop): Dropout(p=0.0, inplace=False)
      )
      (ls1): Identity()
      (drop_path1): Identity()
      (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
      (mlp): Mlp(
        (fc1): Linear(in_features=384, out_features=1536, bias=True)
        (act): GELU(approximate='none')
        (drop1): Dropout(p=0.0, inplace=False)
        (norm): Identity()
        (fc2): Linear(in_features=1536, out_features=384, bias=True)
        (drop2): Dropout(p=0.0, inplace=False)
      )
      (ls2): Identity()
      (drop_path2): Identity()
    )
    (1): Block(
      (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
      (attn): Attention(
        (qkv): Linear(in_features=384, out_features=1152, bias=True)
        (q_norm): Identity()
        (k_norm): Identity()
        (attn_drop): Dropout(p=0.0, inplace=False)
        (norm): Identity()
        (proj): Linear(in_features=384, out_features=384, bias=True)
        (proj_drop): Dropout(p=0.0, inplace=False)
      )
```

```
  (ls1): Identity()
  (drop_path1): Identity()
  (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (mlp): Mlp(
    (fc1): Linear(in_features=384, out_features=1536, bias=True)
    (act): GELU(approximate='none')
    (drop1): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (fc2): Linear(in_features=1536, out_features=384, bias=True)
    (drop2): Dropout(p=0.0, inplace=False)
  )
  (ls2): Identity()
  (drop_path2): Identity()
)
(2): Block(
  (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (attn): Attention(
    (qkv): Linear(in_features=384, out_features=1152, bias=True)
    (q_norm): Identity()
    (k_norm): Identity()
    (attn_drop): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (proj): Linear(in_features=384, out_features=384, bias=True)
    (proj_drop): Dropout(p=0.0, inplace=False)
  )
  (ls1): Identity()
  (drop_path1): Identity()
  (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (mlp): Mlp(
    (fc1): Linear(in_features=384, out_features=1536, bias=True)
    (act): GELU(approximate='none')
    (drop1): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (fc2): Linear(in_features=1536, out_features=384, bias=True)
    (drop2): Dropout(p=0.0, inplace=False)
  )
  (ls2): Identity()
  (drop_path2): Identity()
)
(3): Block(
  (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (attn): Attention(
    (qkv): Linear(in_features=384, out_features=1152, bias=True)
    (q_norm): Identity()
    (k_norm): Identity()
    (attn_drop): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (proj): Linear(in_features=384, out_features=384, bias=True)
```

```
    (proj_drop): Dropout(p=0.0, inplace=False)
  )
  (ls1): Identity()
  (drop_path1): Identity()
  (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (mlp): Mlp(
    (fc1): Linear(in_features=384, out_features=1536, bias=True)
    (act): GELU(approximate='none')
    (drop1): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (fc2): Linear(in_features=1536, out_features=384, bias=True)
    (drop2): Dropout(p=0.0, inplace=False)
  )
  (ls2): Identity()
  (drop_path2): Identity()
)
(4): Block(
  (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (attn): Attention(
    (qkv): Linear(in_features=384, out_features=1152, bias=True)
    (q_norm): Identity()
    (k_norm): Identity()
    (attn_drop): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (proj): Linear(in_features=384, out_features=384, bias=True)
    (proj_drop): Dropout(p=0.0, inplace=False)
  )
  (ls1): Identity()
  (drop_path1): Identity()
  (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (mlp): Mlp(
    (fc1): Linear(in_features=384, out_features=1536, bias=True)
    (act): GELU(approximate='none')
    (drop1): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (fc2): Linear(in_features=1536, out_features=384, bias=True)
    (drop2): Dropout(p=0.0, inplace=False)
  )
  (ls2): Identity()
  (drop_path2): Identity()
)
(5): Block(
  (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (attn): Attention(
    (qkv): Linear(in_features=384, out_features=1152, bias=True)
    (q_norm): Identity()
    (k_norm): Identity()
    (attn_drop): Dropout(p=0.0, inplace=False)
```

```
      (norm): Identity()
      (proj): Linear(in_features=384, out_features=384, bias=True)
      (proj_drop): Dropout(p=0.0, inplace=False)
    )
    (ls1): Identity()
    (drop_path1): Identity()
    (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
    (mlp): Mlp(
      (fc1): Linear(in_features=384, out_features=1536, bias=True)
      (act): GELU(approximate='none')
      (drop1): Dropout(p=0.0, inplace=False)
      (norm): Identity()
      (fc2): Linear(in_features=1536, out_features=384, bias=True)
      (drop2): Dropout(p=0.0, inplace=False)
    )
    (ls2): Identity()
    (drop_path2): Identity()
  )
  (6): Block(
    (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
    (attn): Attention(
      (qkv): Linear(in_features=384, out_features=1152, bias=True)
      (q_norm): Identity()
      (k_norm): Identity()
      (attn_drop): Dropout(p=0.0, inplace=False)
      (norm): Identity()
      (proj): Linear(in_features=384, out_features=384, bias=True)
      (proj_drop): Dropout(p=0.0, inplace=False)
    )
    (ls1): Identity()
    (drop_path1): Identity()
    (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
    (mlp): Mlp(
      (fc1): Linear(in_features=384, out_features=1536, bias=True)
      (act): GELU(approximate='none')
      (drop1): Dropout(p=0.0, inplace=False)
      (norm): Identity()
      (fc2): Linear(in_features=1536, out_features=384, bias=True)
      (drop2): Dropout(p=0.0, inplace=False)
    )
    (ls2): Identity()
    (drop_path2): Identity()
  )
  (7): Block(
    (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
    (attn): Attention(
      (qkv): Linear(in_features=384, out_features=1152, bias=True)
      (q_norm): Identity()
```

```
      (k_norm): Identity()
      (attn_drop): Dropout(p=0.0, inplace=False)
      (norm): Identity()
      (proj): Linear(in_features=384, out_features=384, bias=True)
      (proj_drop): Dropout(p=0.0, inplace=False)
    )
    (ls1): Identity()
    (drop_path1): Identity()
    (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
    (mlp): Mlp(
      (fc1): Linear(in_features=384, out_features=1536, bias=True)
      (act): GELU(approximate='none')
      (drop1): Dropout(p=0.0, inplace=False)
      (norm): Identity()
      (fc2): Linear(in_features=1536, out_features=384, bias=True)
      (drop2): Dropout(p=0.0, inplace=False)
    )
    (ls2): Identity()
    (drop_path2): Identity()
  )
  (8): Block(
    (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
    (attn): Attention(
      (qkv): Linear(in_features=384, out_features=1152, bias=True)
      (q_norm): Identity()
      (k_norm): Identity()
      (attn_drop): Dropout(p=0.0, inplace=False)
      (norm): Identity()
      (proj): Linear(in_features=384, out_features=384, bias=True)
      (proj_drop): Dropout(p=0.0, inplace=False)
    )
    (ls1): Identity()
    (drop_path1): Identity()
    (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
    (mlp): Mlp(
      (fc1): Linear(in_features=384, out_features=1536, bias=True)
      (act): GELU(approximate='none')
      (drop1): Dropout(p=0.0, inplace=False)
      (norm): Identity()
      (fc2): Linear(in_features=1536, out_features=384, bias=True)
      (drop2): Dropout(p=0.0, inplace=False)
    )
    (ls2): Identity()
    (drop_path2): Identity()
  )
  (9): Block(
    (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
    (attn): Attention(
```

```
    (qkv): Linear(in_features=384, out_features=1152, bias=True)
    (q_norm): Identity()
    (k_norm): Identity()
    (attn_drop): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (proj): Linear(in_features=384, out_features=384, bias=True)
    (proj_drop): Dropout(p=0.0, inplace=False)
  )
  (ls1): Identity()
  (drop_path1): Identity()
  (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (mlp): Mlp(
    (fc1): Linear(in_features=384, out_features=1536, bias=True)
    (act): GELU(approximate='none')
    (drop1): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (fc2): Linear(in_features=1536, out_features=384, bias=True)
    (drop2): Dropout(p=0.0, inplace=False)
  )
  (ls2): Identity()
  (drop_path2): Identity()
)
(10): Block(
  (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (attn): Attention(
    (qkv): Linear(in_features=384, out_features=1152, bias=True)
    (q_norm): Identity()
    (k_norm): Identity()
    (attn_drop): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (proj): Linear(in_features=384, out_features=384, bias=True)
    (proj_drop): Dropout(p=0.0, inplace=False)
  )
  (ls1): Identity()
  (drop_path1): Identity()
  (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
  (mlp): Mlp(
    (fc1): Linear(in_features=384, out_features=1536, bias=True)
    (act): GELU(approximate='none')
    (drop1): Dropout(p=0.0, inplace=False)
    (norm): Identity()
    (fc2): Linear(in_features=1536, out_features=384, bias=True)
    (drop2): Dropout(p=0.0, inplace=False)
  )
  (ls2): Identity()
  (drop_path2): Identity()
)
(11): Block(
```

```
        (norm1): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
        (attn): Attention(
          (qkv): Linear(in_features=384, out_features=1152, bias=True)
          (q_norm): Identity()
          (k_norm): Identity()
          (attn_drop): Dropout(p=0.0, inplace=False)
          (norm): Identity()
          (proj): Linear(in_features=384, out_features=384, bias=True)
          (proj_drop): Dropout(p=0.0, inplace=False)
        )
        (ls1): Identity()
        (drop_path1): Identity()
        (norm2): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
        (mlp): Mlp(
          (fc1): Linear(in_features=384, out_features=1536, bias=True)
          (act): GELU(approximate='none')
          (drop1): Dropout(p=0.0, inplace=False)
          (norm): Identity()
          (fc2): Linear(in_features=1536, out_features=384, bias=True)
          (drop2): Dropout(p=0.0, inplace=False)
        )
        (ls2): Identity()
        (drop_path2): Identity()
      )
    )
    (norm): LayerNorm((384,), eps=1e-06, elementwise_affine=True)
    (fc_norm): Identity()
    (head_drop): Dropout(p=0.0, inplace=False)
    (head): Linear(in_features=384, out_features=10, bias=True)
)
```

17 STEP 14: Define Loss Function and Optimizer

[24]:

```
print("\n" + "="*80)
print("STEP 14: DEFINING LOSS FUNCTION & OPTIMIZER")
print("="*80)

LABEL_SMOOTHING = 0.1

criterion = nn.CrossEntropyLoss(label_smoothing=LABEL_SMOOTHING)

print("\n[LOSS FUNCTION]")
print("• Type                : CrossEntropyLoss")
print(f"• Label smoothing     : {LABEL_SMOOTHING}")
print("• Purpose             : Prevent overconfident predictions")
print("• Benefit             : Better generalization for Transformers")
```

```
================================================================================
STEP 14: DEFINING LOSS FUNCTION & OPTIMIZER
================================================================================

[LOSS FUNCTION]
• Type                : CrossEntropyLoss
• Label smoothing     : 0.1
• Purpose             : Prevent overconfident predictions
• Benefit             : Better generalization for Transformers
```

Optimizer (AdamW)

[25]:
```
LEARNING_RATE = 3e-4
WEIGHT_DECAY = 0.05

optimizer = optim.AdamW(
    model.parameters(),
    lr=LEARNING_RATE,
    weight_decay=WEIGHT_DECAY
)

print("\n[OPTIMIZER]")
print("• Optimizer         : AdamW")
print(f"• Learning rate     : {LEARNING_RATE}")
print(f"• Weight decay      : {WEIGHT_DECAY}")
print("• Why AdamW?        : Correct decoupled weight decay (better for ViT)")
```

```
[OPTIMIZER]
• Optimizer           : AdamW
• Learning rate       : 0.0003
• Weight decay        : 0.05
• Why AdamW?          : Correct decoupled weight decay (better for ViT)
```

[26]:
```
print("\n[OPTIMIZER SANITY CHECK]")
print(f"• Number of parameter groups: {len(optimizer.param_groups)}")
print("Expected: 1 parameter group")

if len(optimizer.param_groups) == 1:
    print("  Optimizer correctly attached to model parameters")
else:
    print("  Unexpected optimizer configuration")
```

```
[OPTIMIZER SANITY CHECK]
• Number of parameter groups: 1
Expected: 1 parameter group
  Optimizer correctly attached to model parameters
```

18 STEP 15: Define Learning Rate Scheduler

Controls how fast the model learns over time.

19 Cosine Annealing Scheduler

[27]:
```
print("\n" + "="*80)
print("STEP 15: DEFINING LEARNING RATE SCHEDULER")
print("="*80)

MAX_EPOCHS = 100  # upper bound (early stopping will stop earlier)

scheduler = optim.lr_scheduler.CosineAnnealingLR(
    optimizer,
    T_max=MAX_EPOCHS
)

print("\n[LEARNING RATE SCHEDULER]")
print("• Scheduler type    : CosineAnnealingLR")
print(f"• Maximum epochs    : {MAX_EPOCHS}")
print("• LR behavior       : Gradually decays from initial LR to near zero")
print("• Why cosine?       : Smooth decay improves transformer convergence")
```

```
================================================================================
STEP 15: DEFINING LEARNING RATE SCHEDULER
================================================================================

[LEARNING RATE SCHEDULER]
• Scheduler type    : CosineAnnealingLR
• Maximum epochs    : 100
• LR behavior       : Gradually decays from initial LR to near zero
• Why cosine?       : Smooth decay improves transformer convergence
```

20 STEP 16: Define Training Function

[28]:
```
print("\n" + "="*80)
print("STEP 16: DEFINING TRAINING FUNCTION (MixUp + CutMix)")
print("="*80)

def mixup_data(x, y, alpha=1.0):
    lam = np.random.beta(alpha, alpha)
    batch_size = x.size(0)
    index = torch.randperm(batch_size).to(x.device)

    mixed_x = lam * x + (1 - lam) * x[index]
    y_a, y_b = y, y[index]
```

```
    return mixed_x, y_a, y_b, lam


def cutmix_data(x, y, alpha=1.0):
    lam = np.random.beta(alpha, alpha)
    batch_size, _, H, W = x.size()
    index = torch.randperm(batch_size).to(x.device)

    cx = np.random.randint(W)
    cy = np.random.randint(H)
    w = int(W * np.sqrt(1 - lam))
    h = int(H * np.sqrt(1 - lam))

    x1 = np.clip(cx - w // 2, 0, W)
    x2 = np.clip(cx + w // 2, 0, W)
    y1 = np.clip(cy - h // 2, 0, H)
    y2 = np.clip(cy + h // 2, 0, H)

    x[:, :, y1:y2, x1:x2] = x[index, :, y1:y2, x1:x2]
    lam = 1 - ((x2 - x1) * (y2 - y1) / (W * H))

    y_a, y_b = y, y[index]
    return x, y_a, y_b, lam
```

```
================================================================================
STEP 16: DEFINING TRAINING FUNCTION (MixUp + CutMix)
================================================================================
```

[29]:
```
def train_one_epoch(model, loader, optimizer, criterion, device, epoch):
    model.train()

    running_loss = 0.0
    correct = 0
    total = 0

    print("\n" + "-"*60)
    print(f"Training Epoch {epoch}")
    print("-"*60)

    for batch_idx, (images, labels) in enumerate(loader):
        images, labels = images.to(device), labels.to(device)

        # Randomly choose MixUp or CutMix
        use_cutmix = np.random.rand() < 0.5

        if use_cutmix:
```

```
            images, y_a, y_b, lam = cutmix_data(images, labels)
            aug_type = "CutMix"
        else:
            images, y_a, y_b, lam = mixup_data(images, labels)
            aug_type = "MixUp"

        optimizer.zero_grad()
        outputs = model(images)

        loss = lam * criterion(outputs, y_a) + (1 - lam) * criterion(outputs,␣
↪y_b)
        loss.backward()
        optimizer.step()

        running_loss += loss.item()
        _, predicted = outputs.max(1)

        total += labels.size(0)
        correct += (lam * predicted.eq(y_a).sum().item() +
                    (1 - lam) * predicted.eq(y_b).sum().item())

        if batch_idx == 0:
            print(f"First batch augmentation used: {aug_type}")
            print(f"• Batch image shape: {images.shape}")
            print(f"• Batch label shape: {labels.shape}")
            print("Expected image shape: (batch_size, 3, 32, 32)")

    epoch_loss = running_loss / len(loader)
    epoch_acc = 100. * correct / total

    print(f"Epoch {epoch} Training Loss: {epoch_loss:.4f}")
    print(f"Epoch {epoch} Training Accuracy: {epoch_acc:.2f}%")

    return epoch_loss, epoch_acc
```

21 STEP 17: Define Evaluation Function

[30]:
```
print("\n" + "="*80)
print("STEP 17: DEFINING EVALUATION FUNCTION")
print("="*80)

def evaluate(model, loader, criterion, device, split_name="Validation"):
    model.eval()

    running_loss = 0.0
    correct = 0
```

```
    total = 0

    all_preds = []
    all_labels = []

    print(f"\nEvaluating on {split_name} set...")

    with torch.no_grad():
        for images, labels in loader:
            images, labels = images.to(device), labels.to(device)

            outputs = model(images)
            loss = criterion(outputs, labels)

            running_loss += loss.item()
            _, predicted = outputs.max(1)

            total += labels.size(0)
            correct += predicted.eq(labels).sum().item()

            all_preds.extend(predicted.cpu().numpy())
            all_labels.extend(labels.cpu().numpy())

    avg_loss = running_loss / len(loader)
    acc = 100. * correct / total

    print(f"{split_name} Loss     : {avg_loss:.4f}")
    print(f"{split_name} Accuracy : {acc:.2f}%")

    return avg_loss, acc, all_labels, all_preds
```

```
================================================================================
STEP 17: DEFINING EVALUATION FUNCTION
================================================================================
```

22 STEP 18: Define Early Stopping Logic

[31]:
```
print("\n" + "="*80)
print("STEP 18: DEFINING EARLY STOPPING")
print("="*80)

class EarlyStopping:
    def __init__(self, patience=8, min_delta=0.0):
        self.patience = patience
        self.min_delta = min_delta
        self.best_acc = 0.0
```

```
        self.counter = 0
        self.should_stop = False

        print("\n[EARLY STOPPING CONFIGURATION]")
        print(f"• Patience    : {patience} epochs")
        print(f"• Min delta   : {min_delta}")
        print("• Monitored   : Validation accuracy")

    def step(self, val_acc):
        if val_acc > self.best_acc + self.min_delta:
            print(f" Validation accuracy improved: {self.best_acc:.2f}% →
 ↪{val_acc:.2f}%")
            self.best_acc = val_acc
            self.counter = 0
        else:
            self.counter += 1
            print(f" No improvement. Counter: {self.counter}/{self.patience}")

        if self.counter >= self.patience:
            self.should_stop = True
            print("Early stopping triggered")

        return self.should_stop
```

```
================================================================================
STEP 18: DEFINING EARLY STOPPING
================================================================================
```

23 STEP 19: Two-Stage Fine-Tuning

[32]:
```
print("\n" + "="*80)
print("STEP 19: TWO-STAGE FINE-TUNING")
print("="*80)

print("\n[STAGE 1: FREEZING BACKBONE]")
for name, param in model.named_parameters():
    if "head" not in name:
        param.requires_grad = False

trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad)

print(f"• Trainable parameters after freezing: {trainable_params:,}")
print("• Only classification head is trainable")
```

```
================================================================================
```

```
STEP 19: TWO-STAGE FINE-TUNING
================================================================================

[STAGE 1: FREEZING BACKBONE]
• Trainable parameters after freezing: 3,850
• Only classification head is trainable
```

[33]:

```python
optimizer = optim.AdamW(
    filter(lambda p: p.requires_grad, model.parameters()),
    lr=LEARNING_RATE,
    weight_decay=WEIGHT_DECAY
)

print("Optimizer reconfigured for Stage 1")
```

```
Optimizer reconfigured for Stage 1
```

[34]:

```python
def unfreeze_model(model, n_blocks=12):
    if n_blocks >= 12:
        for param in model.parameters():
            param.requires_grad = True
        print("\n[STAGE 2: UNFREEZING FULL MODEL]")
    else:
        # Freeze all parameters first
        for param in model.parameters():
            param.requires_grad = False

        # Unfreeze the classification head
        for param in model.heads.parameters():
            param.requires_grad = True

        # Unfreeze the last n_blocks of the encoder
        if n_blocks > 0:
            layers = list(model.encoder.layers)
            for layer in layers[-n_blocks:]:
                for param in layer.parameters():
                    param.requires_grad = True

        print(f"\n[STAGE 2: UNFREEZING LAST {n_blocks} BLOCKS + HEAD]")

    trainable = sum(p.numel() for p in model.parameters() if p.requires_grad)
    print(f"• Trainable parameters now: {trainable:,}")
    if n_blocks >= 12:
        print("• Full fine-tuning enabled")
```

24 STEP 20: Create Output Directory

[35]:
```
print("\n" + "="*80)
print("STEP 20: CREATING OUTPUT DIRECTORY")
print("="*80)

OUTPUT_DIR = Path("vit_cifar10_experiment")
OUTPUT_DIR.mkdir(exist_ok=True)

print(f"• Output directory created at: {OUTPUT_DIR.resolve()}")
```

```
================================================================================
STEP 20: CREATING OUTPUT DIRECTORY
================================================================================
• Output directory created at:
/home/onkar/.jackt/sha/vlab/exp2_vit/vit_cifar10_experiment
```

[36]:
```
print("\n" + "="*80)
print("STEP 21: STARTING TRAINING")
print("="*80)

EPOCHS_STAGE1 = 10      # head training
EPOCHS_STAGE2 = 90      # full fine-tuning (upper bound)

early_stopper = EarlyStopping(patience=8)

history = {
    "train_loss": [],
    "train_acc": [],
    "val_loss": [],
    "val_acc": []
}
```

```
================================================================================
STEP 21: STARTING TRAINING
================================================================================

[EARLY STOPPING CONFIGURATION]
• Patience   : 8 epochs
• Min delta  : 0.0
• Monitored  : Validation accuracy
```

[37]:
```
print("\n" + "="*60)
print("STAGE 1 TRAINING: CLASSIFIER HEAD ONLY")
print("="*60)
```

```python
for epoch in range(1, EPOCHS_STAGE1 + 1):
    train_loss, train_acc = train_one_epoch(
        model, train_loader, optimizer, criterion, DEVICE, epoch
    )

    val_loss, val_acc, _, _ = evaluate(
        model, val_loader, criterion, DEVICE, split_name="Validation"
    )

    history["train_loss"].append(train_loss)
    history["train_acc"].append(train_acc)
    history["val_loss"].append(val_loss)
    history["val_acc"].append(val_acc)
```

```
============================================================
STAGE 1 TRAINING: CLASSIFIER HEAD ONLY
============================================================

------------------------------------------------------------
Training Epoch 1
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 1 Training Loss: 2.2682
Epoch 1 Training Accuracy: 25.63%

Evaluating on Validation set…
Validation Loss     : 1.5174
Validation Accuracy : 55.62%

------------------------------------------------------------
Training Epoch 2
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 2 Training Loss: 1.8594
Epoch 2 Training Accuracy: 41.58%

Evaluating on Validation set…
Validation Loss     : 1.2533
Validation Accuracy : 67.54%

------------------------------------------------------------
```

```
Training Epoch 3
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 3 Training Loss: 1.7678
Epoch 3 Training Accuracy: 46.08%

Evaluating on Validation set…
Validation Loss     : 1.1710
Validation Accuracy : 72.02%

------------------------------------------------------------
Training Epoch 4
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 4 Training Loss: 1.7490
Epoch 4 Training Accuracy: 47.31%

Evaluating on Validation set…
Validation Loss     : 1.1260
Validation Accuracy : 73.98%

------------------------------------------------------------
Training Epoch 5
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 5 Training Loss: 1.7498
Epoch 5 Training Accuracy: 47.25%

Evaluating on Validation set…
Validation Loss     : 1.1148
Validation Accuracy : 74.80%

------------------------------------------------------------
Training Epoch 6
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
```

```
Epoch 6 Training Loss: 1.7451
Epoch 6 Training Accuracy: 47.29%

Evaluating on Validation set…
Validation Loss     : 1.1045
Validation Accuracy : 75.38%

------------------------------------------------------------
Training Epoch 7
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 7 Training Loss: 1.7098
Epoch 7 Training Accuracy: 49.13%

Evaluating on Validation set…
Validation Loss     : 1.0815
Validation Accuracy : 76.48%

------------------------------------------------------------
Training Epoch 8
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 8 Training Loss: 1.7097
Epoch 8 Training Accuracy: 49.10%

Evaluating on Validation set…
Validation Loss     : 1.0818
Validation Accuracy : 76.66%

------------------------------------------------------------
Training Epoch 9
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 9 Training Loss: 1.6966
Epoch 9 Training Accuracy: 49.64%

Evaluating on Validation set…
Validation Loss     : 1.0686
Validation Accuracy : 77.56%
```

```
------------------------------------------------------------
Training Epoch 10
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 10 Training Loss: 1.7055
Epoch 10 Training Accuracy: 49.35%

Evaluating on Validation set…
Validation Loss     : 1.0750
Validation Accuracy : 76.44%
```

25 Unfreeze Model for Stage 2

[38]:
```
NUM_BLOCKS_TO_UNFREEZE = 12
```

[39]:
```
unfreeze_model(model, NUM_BLOCKS_TO_UNFREEZE)

optimizer = optim.AdamW(
    model.parameters(),
    lr=LEARNING_RATE / 10,    # smaller LR for fine-tuning
    weight_decay=WEIGHT_DECAY
)

scheduler = optim.lr_scheduler.CosineAnnealingLR(
    optimizer, T_max=EPOCHS_STAGE2
)

print(" Optimizer and scheduler reconfigured for Stage 2")
```

```
[STAGE 2: UNFREEZING FULL MODEL]
• Trainable parameters now: 21,379,210
• Full fine-tuning enabled
 Optimizer and scheduler reconfigured for Stage 2
```

26 Stage 2 Training Loop (Full Fine-Tuning)

[40]:
```
print("\n" + "="*60)
print("STAGE 2 TRAINING: FULL MODEL FINE-TUNING")
print("="*60)

for epoch in range(1, EPOCHS_STAGE2 + 1):
```

```python
    train_loss, train_acc = train_one_epoch(
        model, train_loader, optimizer, criterion, DEVICE, epoch
    )

    val_loss, val_acc, _, _ = evaluate(
        model, val_loader, criterion, DEVICE, split_name="Validation"
    )

    scheduler.step()

    history["train_loss"].append(train_loss)
    history["train_acc"].append(train_acc)
    history["val_loss"].append(val_loss)
    history["val_acc"].append(val_acc)

    if early_stopper.step(val_acc):
        print(f"\nTraining stopped early at epoch {epoch}")
        break
```

```
============================================================
STAGE 2 TRAINING: FULL MODEL FINE-TUNING
============================================================

------------------------------------------------------------
Training Epoch 1
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 1 Training Loss: 1.5969
Epoch 1 Training Accuracy: 54.91%

Evaluating on Validation set…
Validation Loss     : 0.8358
Validation Accuracy : 88.82%
  Validation accuracy improved: 0.00% → 88.82%

------------------------------------------------------------
Training Epoch 2
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 2 Training Loss: 1.5109
Epoch 2 Training Accuracy: 59.44%
```

```
Evaluating on Validation set…
Validation Loss     : 0.8063
Validation Accuracy : 90.10%
 Validation accuracy improved: 88.82% → 90.10%

------------------------------------------------------------
Training Epoch 3
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 3 Training Loss: 1.4261
Epoch 3 Training Accuracy: 63.55%

Evaluating on Validation set…
Validation Loss     : 0.7615
Validation Accuracy : 91.44%
 Validation accuracy improved: 90.10% → 91.44%

------------------------------------------------------------
Training Epoch 4
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 4 Training Loss: 1.4265
Epoch 4 Training Accuracy: 63.31%

Evaluating on Validation set…
Validation Loss     : 0.7535
Validation Accuracy : 92.60%
 Validation accuracy improved: 91.44% → 92.60%

------------------------------------------------------------
Training Epoch 5
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 5 Training Loss: 1.4128
Epoch 5 Training Accuracy: 63.83%

Evaluating on Validation set…
Validation Loss     : 0.7626
```

```
Validation Accuracy : 92.00%
 No improvement. Counter: 1/8

------------------------------------------------------------
Training Epoch 6
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 6 Training Loss: 1.4342
Epoch 6 Training Accuracy: 61.73%

Evaluating on Validation set…
Validation Loss     : 0.7232
Validation Accuracy : 92.82%
 Validation accuracy improved: 92.60% → 92.82%

------------------------------------------------------------
Training Epoch 7
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 7 Training Loss: 1.3316
Epoch 7 Training Accuracy: 66.92%

Evaluating on Validation set…
Validation Loss     : 0.7111
Validation Accuracy : 93.28%
 Validation accuracy improved: 92.82% → 93.28%

------------------------------------------------------------
Training Epoch 8
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 8 Training Loss: 1.3065
Epoch 8 Training Accuracy: 68.54%

Evaluating on Validation set…
Validation Loss     : 0.6956
Validation Accuracy : 93.90%
 Validation accuracy improved: 93.28% → 93.90%
```

```
------------------------------------------------------------
Training Epoch 9
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 9 Training Loss: 1.3324
Epoch 9 Training Accuracy: 66.31%

Evaluating on Validation set…
Validation Loss     : 0.6903
Validation Accuracy : 93.86%
 No improvement. Counter: 1/8

------------------------------------------------------------
Training Epoch 10
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 10 Training Loss: 1.3292
Epoch 10 Training Accuracy: 67.59%

Evaluating on Validation set…
Validation Loss     : 0.7070
Validation Accuracy : 93.52%
 No improvement. Counter: 2/8

------------------------------------------------------------
Training Epoch 11
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 11 Training Loss: 1.3032
Epoch 11 Training Accuracy: 67.70%

Evaluating on Validation set…
Validation Loss     : 0.6703
Validation Accuracy : 93.72%
 No improvement. Counter: 3/8

------------------------------------------------------------
Training Epoch 12
------------------------------------------------------------
```

```
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 12 Training Loss: 1.3046
Epoch 12 Training Accuracy: 68.45%

Evaluating on Validation set…
Validation Loss     : 0.6735
Validation Accuracy : 93.94%
 Validation accuracy improved: 93.90% → 93.94%

------------------------------------------------------------
Training Epoch 13
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 13 Training Loss: 1.2779
Epoch 13 Training Accuracy: 69.01%

Evaluating on Validation set…
Validation Loss     : 0.6818
Validation Accuracy : 94.24%
 Validation accuracy improved: 93.94% → 94.24%

------------------------------------------------------------
Training Epoch 14
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 14 Training Loss: 1.2904
Epoch 14 Training Accuracy: 68.98%

Evaluating on Validation set…
Validation Loss     : 0.6720
Validation Accuracy : 94.28%
 Validation accuracy improved: 94.24% → 94.28%

------------------------------------------------------------
Training Epoch 15
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
```

```
Expected image shape: (batch_size, 3, 32, 32)
Epoch 15 Training Loss: 1.2490
Epoch 15 Training Accuracy: 70.84%

Evaluating on Validation set…
Validation Loss     : 0.6782
Validation Accuracy : 93.82%
 No improvement. Counter: 1/8

------------------------------------------------------------
Training Epoch 16
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 16 Training Loss: 1.2618
Epoch 16 Training Accuracy: 69.67%

Evaluating on Validation set…
Validation Loss     : 0.6760
Validation Accuracy : 94.50%
 Validation accuracy improved: 94.28% → 94.50%

------------------------------------------------------------
Training Epoch 17
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 17 Training Loss: 1.2660
Epoch 17 Training Accuracy: 69.18%

Evaluating on Validation set…
Validation Loss     : 0.6594
Validation Accuracy : 94.60%
 Validation accuracy improved: 94.50% → 94.60%

------------------------------------------------------------
Training Epoch 18
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 18 Training Loss: 1.2778
Epoch 18 Training Accuracy: 68.98%
```

```
Evaluating on Validation set…
Validation Loss     : 0.6579
Validation Accuracy : 94.34%
 No improvement. Counter: 1/8

------------------------------------------------------------
Training Epoch 19
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 19 Training Loss: 1.2799
Epoch 19 Training Accuracy: 68.78%

Evaluating on Validation set…
Validation Loss     : 0.6563
Validation Accuracy : 94.66%
 Validation accuracy improved: 94.60% → 94.66%

------------------------------------------------------------
Training Epoch 20
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 20 Training Loss: 1.2494
Epoch 20 Training Accuracy: 70.68%

Evaluating on Validation set…
Validation Loss     : 0.6672
Validation Accuracy : 94.60%
 No improvement. Counter: 1/8

------------------------------------------------------------
Training Epoch 21
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 21 Training Loss: 1.2701
Epoch 21 Training Accuracy: 68.84%

Evaluating on Validation set…
Validation Loss     : 0.6715
```

```
Validation Accuracy : 94.46%
 No improvement. Counter: 2/8

------------------------------------------------------------
Training Epoch 22
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 22 Training Loss: 1.2021
Epoch 22 Training Accuracy: 72.30%

Evaluating on Validation set…
Validation Loss     : 0.6534
Validation Accuracy : 94.96%
 Validation accuracy improved: 94.66% → 94.96%

------------------------------------------------------------
Training Epoch 23
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 23 Training Loss: 1.2439
Epoch 23 Training Accuracy: 69.64%

Evaluating on Validation set…
Validation Loss     : 0.6526
Validation Accuracy : 94.90%
 No improvement. Counter: 1/8

------------------------------------------------------------
Training Epoch 24
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 24 Training Loss: 1.2676
Epoch 24 Training Accuracy: 69.13%

Evaluating on Validation set…
Validation Loss     : 0.6689
Validation Accuracy : 94.58%
 No improvement. Counter: 2/8
```

```
------------------------------------------------------------
Training Epoch 25
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 25 Training Loss: 1.2223
Epoch 25 Training Accuracy: 70.59%

Evaluating on Validation set…
Validation Loss     : 0.6548
Validation Accuracy : 95.00%
 Validation accuracy improved: 94.96% → 95.00%

------------------------------------------------------------
Training Epoch 26
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 26 Training Loss: 1.1970
Epoch 26 Training Accuracy: 72.84%

Evaluating on Validation set…
Validation Loss     : 0.6481
Validation Accuracy : 95.06%
 Validation accuracy improved: 95.00% → 95.06%

------------------------------------------------------------
Training Epoch 27
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 27 Training Loss: 1.2253
Epoch 27 Training Accuracy: 70.27%

Evaluating on Validation set…
Validation Loss     : 0.6555
Validation Accuracy : 95.20%
 Validation accuracy improved: 95.06% → 95.20%

------------------------------------------------------------
Training Epoch 28
------------------------------------------------------------
```

```
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 28 Training Loss: 1.2380
Epoch 28 Training Accuracy: 70.17%

Evaluating on Validation set…
Validation Loss     : 0.6689
Validation Accuracy : 94.94%
 No improvement. Counter: 1/8

------------------------------------------------------------
Training Epoch 29
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 29 Training Loss: 1.1820
Epoch 29 Training Accuracy: 73.34%

Evaluating on Validation set…
Validation Loss     : 0.6633
Validation Accuracy : 94.90%
 No improvement. Counter: 2/8

------------------------------------------------------------
Training Epoch 30
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 30 Training Loss: 1.2242
Epoch 30 Training Accuracy: 71.18%

Evaluating on Validation set…
Validation Loss     : 0.6501
Validation Accuracy : 94.76%
 No improvement. Counter: 3/8

------------------------------------------------------------
Training Epoch 31
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
```

```
Expected image shape: (batch_size, 3, 32, 32)
Epoch 31 Training Loss: 1.2186
Epoch 31 Training Accuracy: 70.80%

Evaluating on Validation set…
Validation Loss     : 0.6347
Validation Accuracy : 95.18%
 No improvement. Counter: 4/8

------------------------------------------------------------
Training Epoch 32
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 32 Training Loss: 1.1568
Epoch 32 Training Accuracy: 74.19%

Evaluating on Validation set…
Validation Loss     : 0.6375
Validation Accuracy : 95.18%
 No improvement. Counter: 5/8

------------------------------------------------------------
Training Epoch 33
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 33 Training Loss: 1.2653
Epoch 33 Training Accuracy: 68.56%

Evaluating on Validation set…
Validation Loss     : 0.6671
Validation Accuracy : 95.14%
 No improvement. Counter: 6/8

------------------------------------------------------------
Training Epoch 34
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 34 Training Loss: 1.2057
Epoch 34 Training Accuracy: 71.67%
```

```
Evaluating on Validation set…
Validation Loss     : 0.6415
Validation Accuracy : 95.40%
 Validation accuracy improved: 95.20% → 95.40%

------------------------------------------------------------
Training Epoch 35
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 35 Training Loss: 1.2265
Epoch 35 Training Accuracy: 69.96%

Evaluating on Validation set…
Validation Loss     : 0.6470
Validation Accuracy : 95.32%
 No improvement. Counter: 1/8

------------------------------------------------------------
Training Epoch 36
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 36 Training Loss: 1.2039
Epoch 36 Training Accuracy: 71.49%

Evaluating on Validation set…
Validation Loss     : 0.6412
Validation Accuracy : 95.38%
 No improvement. Counter: 2/8

------------------------------------------------------------
Training Epoch 37
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 37 Training Loss: 1.2132
Epoch 37 Training Accuracy: 70.92%

Evaluating on Validation set…
Validation Loss     : 0.6600
```

```
Validation Accuracy : 95.26%
 No improvement. Counter: 3/8

------------------------------------------------------------
Training Epoch 38
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 38 Training Loss: 1.2183
Epoch 38 Training Accuracy: 69.74%

Evaluating on Validation set…
Validation Loss     : 0.6436
Validation Accuracy : 95.20%
 No improvement. Counter: 4/8

------------------------------------------------------------
Training Epoch 39
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 39 Training Loss: 1.1776
Epoch 39 Training Accuracy: 72.35%

Evaluating on Validation set…
Validation Loss     : 0.6363
Validation Accuracy : 95.44%
 Validation accuracy improved: 95.40% → 95.44%

------------------------------------------------------------
Training Epoch 40
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 40 Training Loss: 1.1505
Epoch 40 Training Accuracy: 74.62%

Evaluating on Validation set…
Validation Loss     : 0.6434
Validation Accuracy : 95.38%
 No improvement. Counter: 1/8
```

```
------------------------------------------------------------
Training Epoch 41
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 41 Training Loss: 1.2084
Epoch 41 Training Accuracy: 70.80%

Evaluating on Validation set…
Validation Loss     : 0.6491
Validation Accuracy : 95.04%
 No improvement. Counter: 2/8

------------------------------------------------------------
Training Epoch 42
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 42 Training Loss: 1.1796
Epoch 42 Training Accuracy: 72.40%

Evaluating on Validation set…
Validation Loss     : 0.6425
Validation Accuracy : 95.38%
 No improvement. Counter: 3/8

------------------------------------------------------------
Training Epoch 43
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 43 Training Loss: 1.1993
Epoch 43 Training Accuracy: 72.05%

Evaluating on Validation set…
Validation Loss     : 0.6356
Validation Accuracy : 95.18%
 No improvement. Counter: 4/8

------------------------------------------------------------
Training Epoch 44
------------------------------------------------------------
```

```
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 44 Training Loss: 1.1787
Epoch 44 Training Accuracy: 72.17%

Evaluating on Validation set…
Validation Loss     : 0.6400
Validation Accuracy : 95.26%
 No improvement. Counter: 5/8

------------------------------------------------------------
Training Epoch 45
------------------------------------------------------------
First batch augmentation used: CutMix
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 45 Training Loss: 1.1811
Epoch 45 Training Accuracy: 72.25%

Evaluating on Validation set…
Validation Loss     : 0.6425
Validation Accuracy : 95.32%
 No improvement. Counter: 6/8

------------------------------------------------------------
Training Epoch 46
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
Expected image shape: (batch_size, 3, 32, 32)
Epoch 46 Training Loss: 1.1839
Epoch 46 Training Accuracy: 72.10%

Evaluating on Validation set…
Validation Loss     : 0.6512
Validation Accuracy : 95.00%
 No improvement. Counter: 7/8

------------------------------------------------------------
Training Epoch 47
------------------------------------------------------------
First batch augmentation used: MixUp
• Batch image shape: torch.Size([128, 3, 32, 32])
• Batch label shape: torch.Size([128])
```

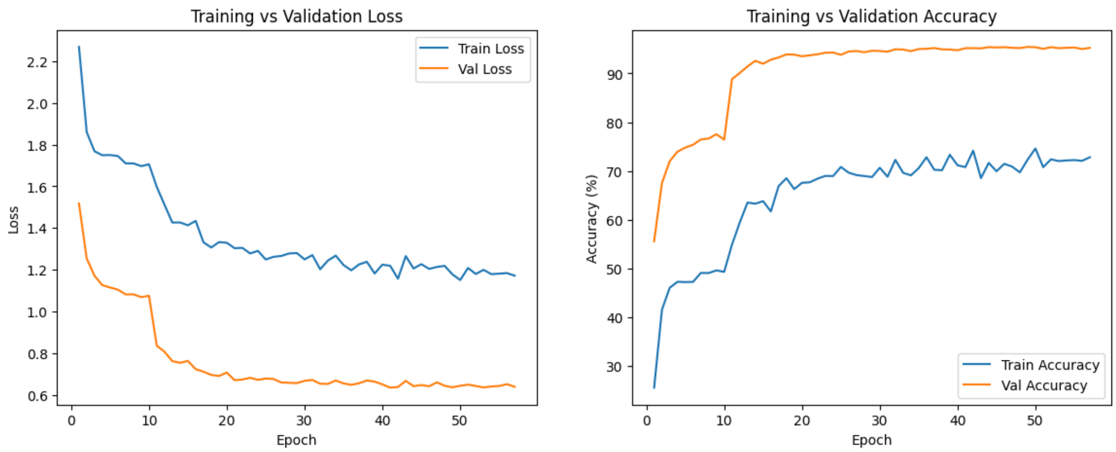
```
Expected image shape: (batch_size, 3, 32, 32)
Epoch 47 Training Loss: 1.1714
Epoch 47 Training Accuracy: 72.83%

Evaluating on Validation set…
Validation Loss      : 0.6387
Validation Accuracy : 95.24%
 No improvement. Counter: 8/8
Early stopping triggered

Training stopped early at epoch 47
```

27 STEP 22: Plot Training Curves (Loss & Accuracy)

[41]:

```
print("\n" + "="*80)
print("STEP 22: PLOTTING TRAINING CURVES")
print("="*80)

epochs = range(1, len(history["train_loss"]) + 1)

plt.figure(figsize=(14, 5))

# Loss
plt.subplot(1, 2, 1)
plt.plot(epochs, history["train_loss"], label="Train Loss")
plt.plot(epochs, history["val_loss"], label="Val Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Training vs Validation Loss")
plt.legend()

# Accuracy
plt.subplot(1, 2, 2)
plt.plot(epochs, history["train_acc"], label="Train Accuracy")
plt.plot(epochs, history["val_acc"], label="Val Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy (%)")
plt.title("Training vs Validation Accuracy")
plt.legend()

plt.show()

print("\nExpected behavior:")
print("• Training loss ↓ steadily")
print("• Validation loss ↓ then stabilizes")
print("• Early stopping prevents divergence")
```

```
================================================================================
STEP 22: PLOTTING TRAINING CURVES
================================================================================
```




```
Expected behavior:
• Training loss ↓ steadily
• Validation loss ↓ then stabilizes
• Early stopping prevents divergence
```

28 STEP 23: Visualize Sample Predictions (Subset Test)

[42]:
```
print("\n" + "="*80)
print("STEP 23: VISUALIZING SAMPLE PREDICTIONS")
print("="*80)

model.eval()
images, labels = next(iter(subset_test_loader))
images, labels = images.to(DEVICE), labels.to(DEVICE)

with torch.no_grad():
    outputs = model(images)
    _, preds = outputs.max(1)

plt.figure(figsize=(15, 4))
for i in range(8):
    img = images[i].cpu()
    img = img * torch.tensor(CIFAR_STD).view(3,1,1) + torch.tensor(CIFAR_MEAN).
  ↪view(3,1,1)
    img = img.clamp(0,1)
```

```python
    plt.subplot(1, 8, i + 1)
    plt.imshow(img.permute(1,2,0))
    plt.title(f"P: {CLASSES[preds[i]]}\nT: {CLASSES[labels[i]]}")
    plt.axis("off")

plt.show()

print("\nInterpretation:")
print("• Correct predictions → confidence in learning")
print("• Wrong predictions → visually ambiguous cases")
```

```
================================================================================
STEP 23: VISUALIZING SAMPLE PREDICTIONS
================================================================================
```


```
Interpretation:
• Correct predictions → confidence in learning
• Wrong predictions → visually ambiguous cases
```

29 STEP 24: Calculate Per-Class Accuracy

[43]:
```python
print("\n" + "="*80)
print("STEP 24: PER-CLASS ACCURACY CALCULATION")
print("="*80)

def per_class_accuracy(labels, preds, num_classes):
    correct = np.zeros(num_classes)
    total = np.zeros(num_classes)

    for l, p in zip(labels, preds):
        total[l] += 1
        if l == p:
            correct[l] += 1

    return 100 * correct / total
```

```
================================================================================
STEP 24: PER-CLASS ACCURACY CALCULATION
================================================================================
```

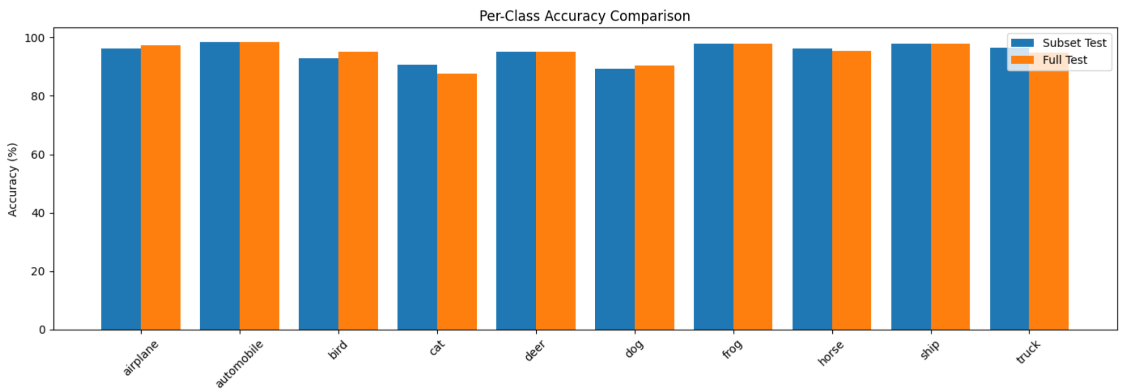
[44]:
```python
_, _, subset_labels, subset_preds = evaluate(
    model, subset_test_loader, criterion, DEVICE, split_name="Subset Test"
)

_, _, full_labels, full_preds = evaluate(
    model, full_test_loader, criterion, DEVICE, split_name="Full Test"
)

subset_acc = per_class_accuracy(subset_labels, subset_preds, NUM_CLASSES)
full_acc = per_class_accuracy(full_labels, full_preds, NUM_CLASSES)
```

```
Evaluating on Subset Test set…
Subset Test Loss     : 0.6529
Subset Test Accuracy : 95.10%

Evaluating on Full Test set…
Full Test Loss     : 0.6469
Full Test Accuracy : 94.98%
```

[45]:
```python
print("\n" + "="*80)
print("STEP 25: PER-CLASS ACCURACY VISUALIZATION")
print("="*80)

x = np.arange(NUM_CLASSES)

plt.figure(figsize=(14, 5))
plt.bar(x - 0.2, subset_acc, width=0.4, label="Subset Test")
plt.bar(x + 0.2, full_acc, width=0.4, label="Full Test")

plt.xticks(x, CLASSES, rotation=45)
plt.ylabel("Accuracy (%)")
plt.title("Per-Class Accuracy Comparison")
plt.legend()
plt.tight_layout()
plt.show()

print("\nInterpretation:")
print("• Subset test is usually slightly higher")
print("• Full test reflects real generalization")
```

```
================================================================================
STEP 25: PER-CLASS ACCURACY VISUALIZATION
```

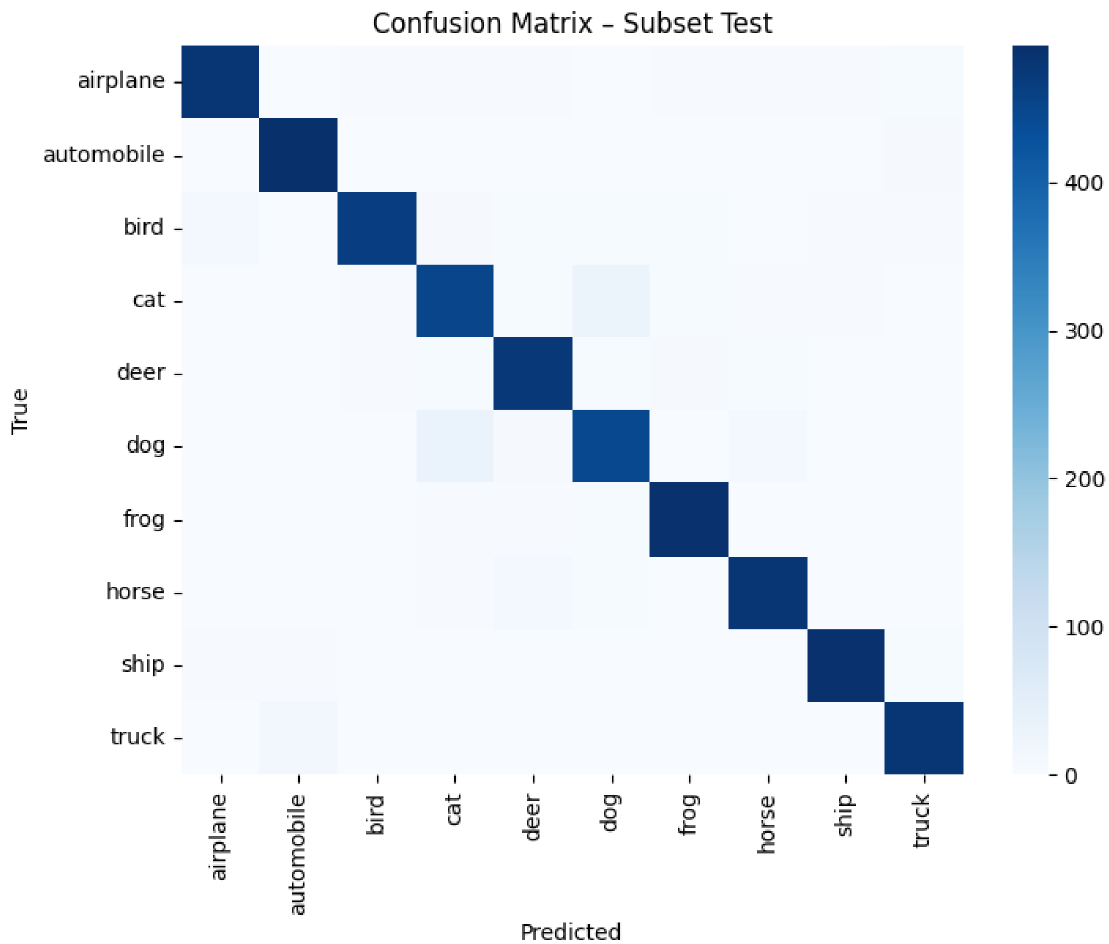
==



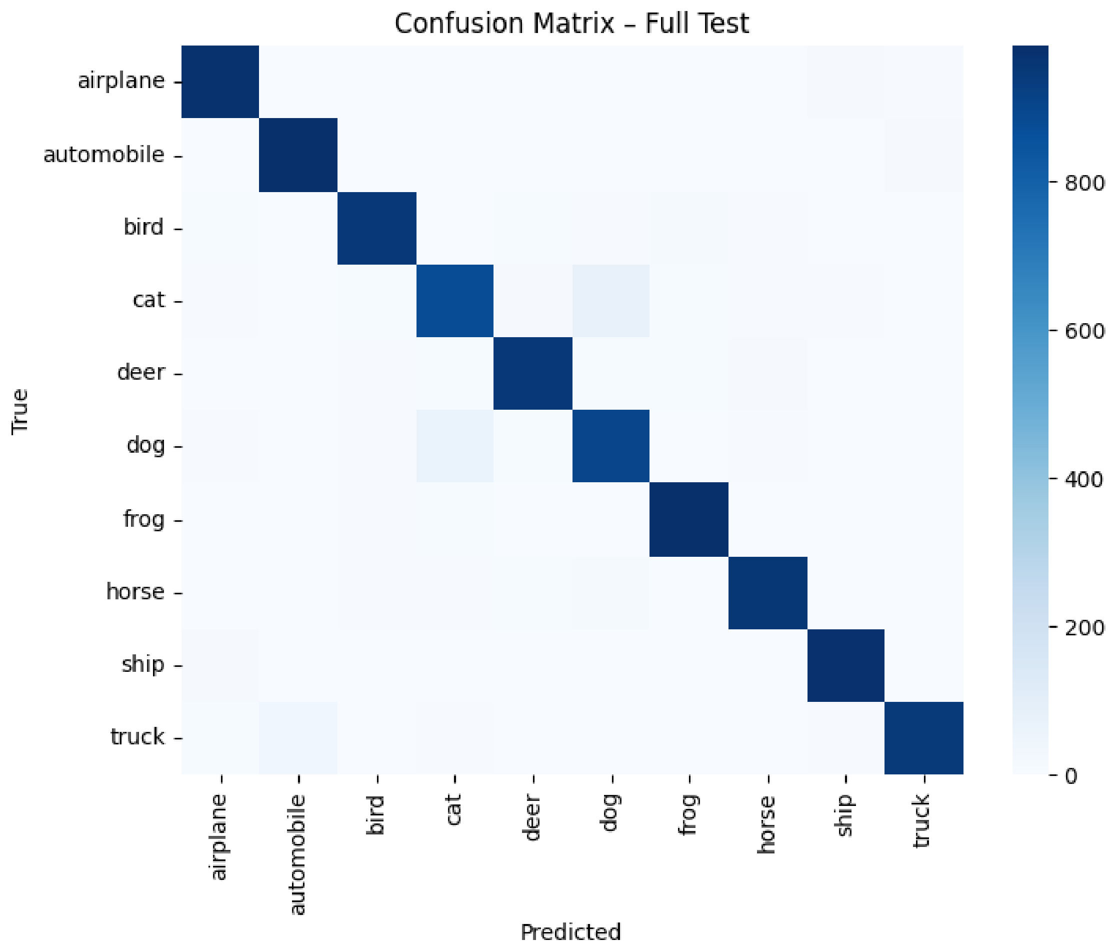
Interpretation:
• Subset test is usually slightly higher
• Full test reflects real generalization

[46]:
```python
print("\n" + "="*80)
print("STEP 26: CONFUSION MATRICES")
print("="*80)

def plot_confusion_matrix(labels, preds, title):
    cm = confusion_matrix(labels, preds)
    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=False, cmap="Blues",
                xticklabels=CLASSES, yticklabels=CLASSES)
    plt.xlabel("Predicted")
    plt.ylabel("True")
    plt.title(title)
    plt.show()

plot_confusion_matrix(subset_labels, subset_preds, "Confusion Matrix - Subset␣
↪Test")
plot_confusion_matrix(full_labels, full_preds, "Confusion Matrix - Full Test")
```

```
================================================================================
STEP 26: CONFUSION MATRICES
================================================================================
```



Confusion Matrix – Subset Test
airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck
True
Predicted
400
300
200
100
0




```
[47]: print("\n" + "="*80)
print("STEP 27: FINAL SUMMARY")
print("="*80)

print("MODEL              : ViT-Small (Patch 8), ImageNet pretrained")
print("DATASET            : CIFAR-10 (class-balanced subset)")
print("TRAIN SAMPLES      : 20,000")
print("VALIDATION SAMPLES : 5,000")
print("TEST (SUBSET)      : 5,000")
print("TEST (FULL)        : 10,000")
print("\nKEY TECHNIQUES USED:")
print("• Transfer learning (ImageNet → CIFAR-10)")
print("• Strong augmentation (MixUp + CutMix)")
print("• Two-stage fine-tuning")
print("• Early stopping")
print("• Cosine learning rate schedule")
```

```python
print("\nWHAT STUDENTS SHOULD REMEMBER:")
print("• ViTs need pretraining")
print("• Patch size matters")
print("• Validation controls training")
print("• Attention replaces convolution")

print("\nEXPERIMENT COMPLETE")
```

```
================================================================================
STEP 27: FINAL SUMMARY
================================================================================
MODEL              : ViT-Small (Patch 8), ImageNet pretrained
DATASET            : CIFAR-10 (class-balanced subset)
TRAIN SAMPLES      : 20,000
VALIDATION SAMPLES : 5,000
TEST (SUBSET)      : 5,000
TEST (FULL)        : 10,000

KEY TECHNIQUES USED:
• Transfer learning (ImageNet → CIFAR-10)
• Strong augmentation (MixUp + CutMix)
• Two-stage fine-tuning
• Early stopping
• Cosine learning rate schedule

WHAT STUDENTS SHOULD REMEMBER:
• ViTs need pretraining
• Patch size matters
• Validation controls training
• Attention replaces convolution

EXPERIMENT COMPLETE
```